

WaveKitchen

A Signal Processing Cooking Game

GAMEPLAY OVERVIEW & EXPERIMENTAL LAB MANUAL · v2.1

FFT · Frequency-Domain Filtering · Superposition · Amplitude & Frequency Tuning
Time Scaling · Time Shifting · AM Modulation · Decimation / Nyquist
LTI Convolution · 4-Part Metric Scoring · Phased Array Acoustic Beam Delivery

1. What This Game Is

In **WaveKitchen**, food is sound. Every ingredient you pick up is an **audio signal**, and every action you take in the kitchen is an authentic, authoritative **signal processing operation**. You do not click a generic button and watch a progress bar. You inspect the ingredient frequency spectrum, discriminate between contaminant noise and harmonic flavor, filter the interference using linear phase tools, combine ingredient waveforms in the mixing bowl, transform amplitude and time scales, and convolve the dish with appliance impulse responses.

Every station is **audible** and **interactive**. You listen to the difference between time scaling and time shifting, watch constructive interference steer phased acoustic beams, and receive objective score breakdowns calculated server-side.

The Golden Rule of WaveKitchen:
The reference dish is produced by evaluating the canonical DSP pipeline on pristine ingredient waveforms using the target parameters. The player dish is evaluated through the exact same mathematical engine. Achieving a high score means matching the underlying mathematics — results cannot be forged client-side.

Station	Kitchen Action	Signal Processing Domain Concept
01. The Pantry	Ingredient Selection & Delivery	Harmonic Synthesis & Stochastic Noise Contamination
02. Filtering Lab	Washing Fresh Produce	FFT, Spectral Masking, Frequency-Domain Low/Bandpass, IFFT
03. Mixing Bowl	Combining Ingredients	Linear Superposition: $m[n] = (1/\sqrt{K}) \sum x_k[n]$
04. Seasoning Lab	Spicing & Flavor Balancing	Amplitude Scaling ($A \cdot x$) & Frequency Scaling ($f \cdot t$)
05. Marinating Lab	Resting & Brining	Time Scaling ($y(t) = x(\alpha t)$) & Time Shifting (Linear Phase Delay)
06. Cooking Lab	Grilling, Frying, Baking, Boiling	LTI Convolution: $y[n] = x[n] * h[n]$ with Cascaded Systems
07. Final Comparison	Tasting & Evaluation	MSE, SNR (dB), Cross-Correlation, Spectral Cosine Similarity
08. Beam Delivery	Phased Array Serving	Acoustic Phased Array Steering, Array Factor $AF(\theta)$, Constructive Beam

2. Ingredient Catalogue & Harmonic Models

WaveKitchen features **18 ingredients** across 5 culinary categories. Produce items arrive washable with high-frequency stochastic noise, whereas bakery, protein, dairy, and pantry items arrive in clean studio conditions.

Ingredient	Category	Voice / Instrument	f ₀ (Hz)	Washable?	Spectral Signature & Harmonics
Lettuce ■	Produce	Shaker	440	Yes (380 Hz)	Band-limited noise with sustained series
Tomato ■	Produce	Kalimba	262	Yes (520 Hz)	Tall harmonic stack decaying at progressive rates
Onion ■	Produce	Oboe	587	Yes (640 Hz)	Bright nasal double-reed upper harmonic spectrum
Cucumber ■	Produce	Piccolo	450	Yes (450 Hz)	High-register agile woodwind tone with flutter
Carrot ■	Produce	Clarinet	500	Yes (500 Hz)	Odd-harmonic dominated cylindrical bore response
Bun ■	Bakery	Marimba	160	No	Resonant wooden bar strike with fast transient decay
Bread ■	Bakery	Marimba	160	No	Tuned woody harmonic bar resonance
Noodles ■	Bakery	Harp	220	No	Cascading plucked nylon string harmonics
Flour ■	Bakery	Organ	130	No	Multi-rank pipe organ steady harmonic series
Beef Patty ■	Protein	Bass Drum	90	No	Fat low-frequency fundamental with click attack
Chicken ■	Protein	Bass Drum	110	No	Punchy low-mid drum transient
Cheese ■	Dairy	Flute	330	No	Pure woodwind tone with soft breath sideband
Egg ■	Protein	Kalimba	260	No	Plucked metallic tine decay
Milk ■	Dairy	Flute	300	No	Smooth sustained acoustic woodwind envelope
Butter ■	Dairy	Bassoon	180	No	Warm reedy low double-reed harmonics
Sugar ■	Pantry	Glockenspiel	700	No	Bright crystalline high metallic bell chime
Salt ■	Pantry	Triangle	800	No	High ringing crystalline steel sustain
Sauce ■	Pantry	Clarinet	240	No	Rich odd-harmonic woodwind resonance

3. Kitchen Stations & Signal Transformations

Station 02 · The Fourier Filtering Machine

Fresh produce gathers stochastic contamination on delivery (white noise, mains hum, high-frequency hiss, burst crackle, stray tones). The player sculpts the frequency spectrum using an 8-band graphic equalizer and parametric tools (low-pass, high-pass, band-pass, notch, de-hum). The cleanliness score balances noise removal against flavor preservation:
Cleanliness = 100 × (0.55 × Removal + 0.45 × Preservation).

Station 03 · Superposition Mixing Bowl

Cleaned ingredient signals enter the bowl, undergoing sample-by-sample linear superposition normalized by the square root of the ingredient count: $\mathbf{m}[n] = (1 / \sqrt{K}) \sum \mathbf{x_k}[n]$. The mixed spectrum displays harmonic interleaving.

Station 04 · Seasoning Lab (Amplitude & Frequency Scaling)

Adjusts signal amplitude (scaling volume/magnitude: $\mathbf{s}[n] = \mathbf{A} \cdot \mathbf{m}[n]$) and frequency characteristics (harmonic distribution tuning) to match the target recipe waveform profile.

Station 05 · Marinating Lab (Time Scaling & Time Shifting)

Implements duration expansion/compression $\mathbf{y(t)} = \mathbf{x(\alpha t)}$. Faster speeds ($\alpha > 1$) compress duration and stretch spectrum; slower speeds ($\alpha < 1$) lengthen duration and compress bandwidth. Fractional time delay $\mathbf{y(t)} = \mathbf{x(t - t_{\Delta})}$ is applied as a linear phase ramp in frequency domain.

Station 06 · Cooking Lab (LTI System Convolution)

Every appliance is an LTI system defined by its impulse response $\mathbf{h[n]}$: $\mathbf{y[n]} = \mathbf{x[n]} * \mathbf{h[n]}$.

- **Grill:** Comb filter with 90-sample discrete echoes (sharp spiky taps).
- **Fry:** Damped exponential noise decay (crackle bursts).
- **Bake:** 2600-tap dark low-passed decay (smooth warming tail).
- **Boil:** Damped 320 Hz resonant bandpass (slow rolling bubbles).

Station 07 · Final Comparison & Score Breakdown

The cooked dish is scored against the reference dish using four objective metrics: Mean Squared Error (MSE), Signal-to-Noise Ratio (SNR dB), Normalised Cross-Correlation (r_{xy}), and Log-Spectral Cosine Similarity. The screen presents a 4-part grade breakdown: **Filtering**, **Mixing**, **Transformation**, and **Cooking**.

Station 08 · Phased Array Acoustic Beam Delivery (NEW)

An 8-element linear speaker array delivers the cooked soundwave to diner tables across the dining hall. By applying progressive phase shifts $\Delta\phi$ across array elements, constructive interference steers the acoustic main lobe to the target radial angle θ : $\Delta\phi = -2\pi (d/\lambda) \sin(\theta)$.
The server evaluates array factor radiation patterns $\mathbf{AF(\theta)}$ and confirms alignment within $\pm 6^\circ$ tolerance.

4. Master Recipe Book

Recipe	Tier / Difficulty	Ingredients	Appliance	Seasoning (Amp, Freq)	Marinate (Scale)
Burger ■	Tier 1 / Easy	Bun, Patty, Cheese, Lettuce, Tomato, Salt	Grill	A = 1.5, f = 0.8	1.25×
Sandwich ■	Tier 1 / Easy	Bread, Chicken, Cheese, Lettuce, Tomato, Salt, Sauce	Grill	A = 1.2, f = 1.1	0.85×
Cake ■	Tier 2 / Medium	Flour, Egg, Butter, Sugar, Milk	Bake	A = 1.8, f = 0.6	1.50×
Noodles ■	Tier 2 / Medium	Noodles, Egg, Chicken, Onion, Salt	Boil	A = 1.3, f = 1.2	1.75×
Chicken Fry ■	Tier 3 / Hard	Chicken, Flour, Egg, Salt, Butter	Fry	A = 2.0, f = 1.4	1.40×